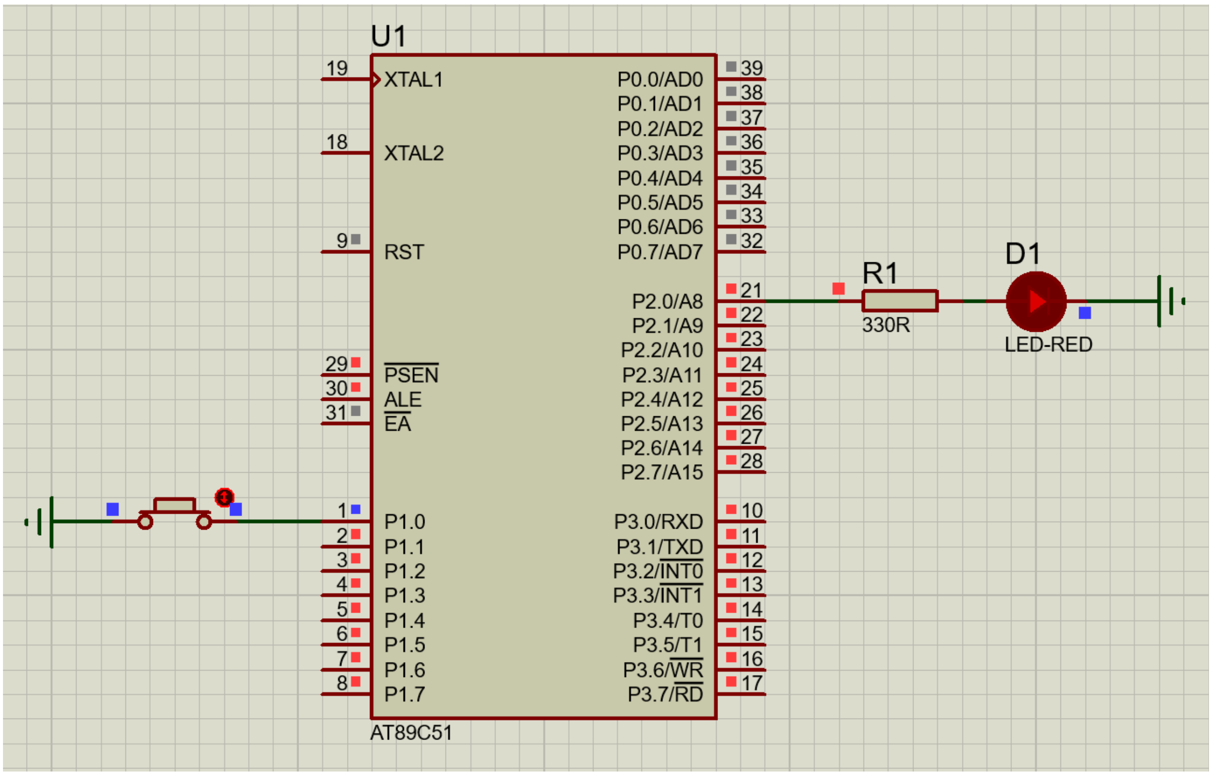
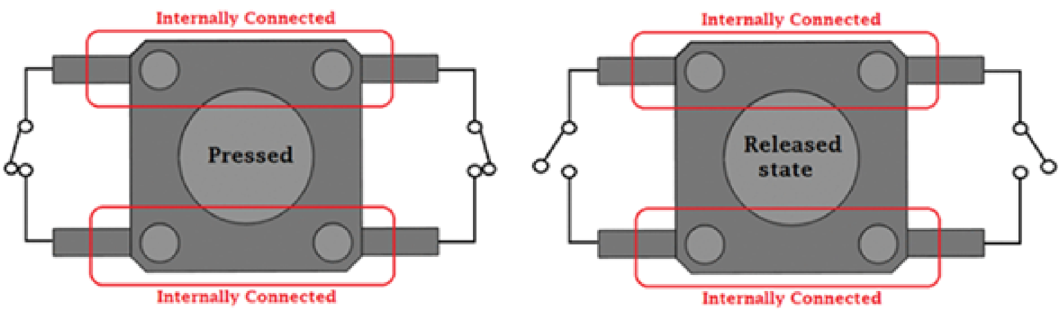
2️⃣ Push Button - Control LED

Aim: To write & execute a program to interface a push button with the 8051 microcontroller & control an LED based on button I/P.





Apparatus Required:

- 8051 microcontroller
- Push button switch
- LEDs
- Resistor (330Ω)
- Breadboard

- Power Supply (5V)
- Keil uVision IDE
- Connecting wires

Theory:

- A push button is an I/P device used to provide digital signals to the microcontroller.
- When pressed, the logic changes (High to Low).
- When released, it returns to its original state.
- **Pull-up Concept**:
  - Port pins of the 8051 have internal pull-up resistors.
  - Normally, the I/P reads High (1).
  - When the button is pressed, it connects to GND → Low (0).

Working:

- Button not pressed → I/P = 1.
- Button pressed → I/P = 0.
- Based on this I/P, the microcontroller controls the LED.

Program Code:

```
#include <reg51.h>

// Define hardware connections (capitalized for constants/hardware
pins)
sbit BUTTON = P1^0;
sbit LED    = P2^0;

void main()
{
    // 1. CRITICAL: Configure the button pin as an INPUT
    BUTTON = 1;

    // 2. Initialize LED to OFF
    LED = 0;

    while(1)
```

```
    {
        // 3. Directly invert the button state to control the LED
        // If BUTTON is 0 (pressed), LED becomes 1 (ON).
        // If BUTTON is 1 (released), LED becomes 0 (OFF).
        LED = ~BUTTON;
    }
}
```

Result: The push button was successfully interfaced with the 8051 microcontroller & the LED responded correctly to button I/P.

Precautions:

- Avoid floating I/Ps.
- Ensure proper wiring of the button.
- Use debounce delay to avoid false triggering.
- Check the polarity of the LED.